

ANDREW (A.J.) SAGER

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Sustainability & GIS Analytics

PROFESSIONAL SUMMARY

Business Analytics M.S. graduate and former Division I athlete with hands-on experience in geospatial machine learning, conservation prioritization, and clean energy program evaluation. Builds GIS-driven decision-support tools for environmental and mission-driven organizations.

FEATURED PROJECT

Climate-Resilient Conservation Prioritization Using GIS and Machine Learning — Independent Project

- Built an end-to-end geospatial ML decision-support framework prioritizing conservation investments across California under climate change, as a reproducible Python pipeline
- Modeled species habitat suitability on a statewide 10 km grid using 2,300+ GBIF occurrence records, WorldClim CMIP6 climate projections, CAL FIRE FRAP wildfire history, CPAD protected areas, elevation, and land cover; compared Random Forest, XGBoost, and Logistic Regression with spatial block cross-validation and removed observation-effort bias, validating results against California's ACE dataset
- Built a multi-criteria prioritization framework (climate resilience, wildfire risk, protection status, connectivity) with least-cost path corridor analysis; formulated a budget-constrained land acquisition problem as a binary integer program (PuLP) and ran Monte Carlo sensitivity and multi-objective portfolio analysis
- Tools: Python, GeoPandas, Rasterio, Shapely, Scikit-learn, XGBoost, NetworkX, PuLP

WORK EXPERIENCE

Business Analyst — *EcoMetricx*

Jan 2026 – Present

- Spearheaded an 82-page program evaluation for an Ohio community energy nonprofit, analyzing 48 programs against a custom KPI framework and applying PCA to identify drivers of clean energy program success
- Conducted 30+ stakeholder interviews with client leadership, community choice aggregators (CCAs), and partner organizations to synthesize program impact findings
- Won a consulting contract with a public-private partnership reducing food waste by connecting food businesses with community organizations
- Produced a 25-year organizational retrospective and an OSU engagement synthesis informing future sustainability strategy; supported RFP processes with a Rent Stabilization Board and City Council on municipal equity-focused analytics engagements

Student Capstone Data Analyst — *Schneider Electric*

Jan 2025 – June 2025

- Built an ML forecasting model across four California housing markets for Schneider Electric; served as project manager and client liaison, aligning outputs to sustainability strategy goals

Financial Planning & Analysis Intern — *Sidoxia, LLC*

May 2023 – Nov 2023

- Conducted ESG alignment and sustainability-impact analysis of major investment portfolios (BlackRock, Blackstone) using R and Excel; built forecasting models supporting portfolios exceeding \$200M

EDUCATION

M.S. Business Analytics

June 2025

UC Irvine, Paul Merage School of Business

GPA: 3.88 | Beta Gamma Sigma Honor Society (Top 20% of class)

B.S. Quantitative Economics

March 2024

UC Irvine

GPA: 3.70 / 3.94

SKILLS

GIS & Spatial: GeoPandas, Rasterio, Shapely, Spatial Analysis, Species Distribution Modeling, Least-Cost Path Analysis, Spatial Optimization

Sustainability & Policy: Clean Energy Program Evaluation, Community Choice Aggregators (CCAs), ESG Analysis, Climate Data Analysis (WorldClim CMIP6), Municipal Analytics

Analytical Methods: Machine Learning (Random Forest, XGBoost, Logistic Regression), Integer Programming (PuLP), Monte Carlo Simulation, PCA, Econometric Modeling, KPI Framework Design

Programming & Tools: Python (Pandas, NumPy, Scikit-learn, NetworkX), R (tidyverse, ggplot2), SQL, Excel, Tableau, Power BI, Git/GitHub

ADDITIONAL EXPERIENCE

- Ecological Data Science Research Assistant, UC Irvine — field research in Southern California Coastal Sage Scrub ecosystems, applying regression modeling in R and Python to study plant-herbivore dynamics
- Division I Athlete, Track & Cross Country, UC Irvine (2020–2025), while maintaining a 3.88 graduate GPA